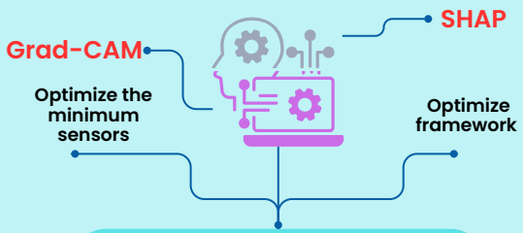
Objective

Explainable AI Model



SUITABLE FOR FINGERS CLASSIFICATION

SUITABLE FOR WIDE RANGE OF USERS

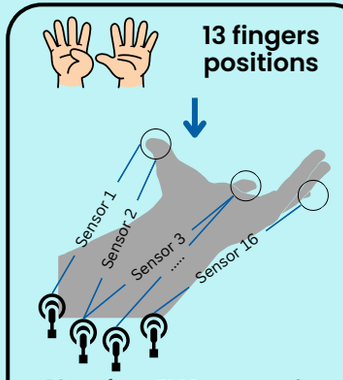
Leave-One Subject Out Validation (LOSO)

SIMPLE MODEL SUITABLE FOR REAL-TIME CLASSIFICATION

Use simple CNN structure

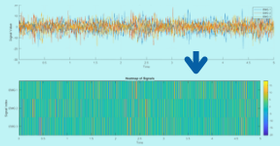
Methodology

Ninapro DB5 Dataset



16 surface EMG sensors from Myo-armbands

Deep Learning Framework



Matrix from sEMG signal channels

CNN Model

Grad-CAM

Feature extraction (500ms window)

IQR | RMS

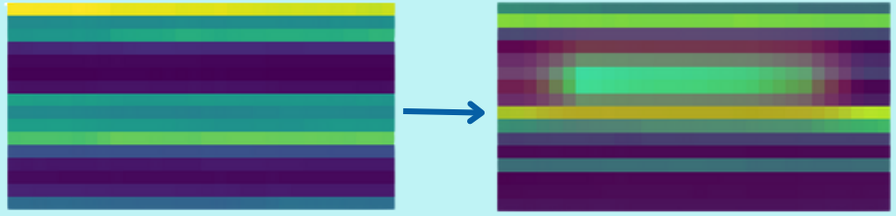
Entropy | ...

Machine Learning

Random Forest | Support Vector Machine

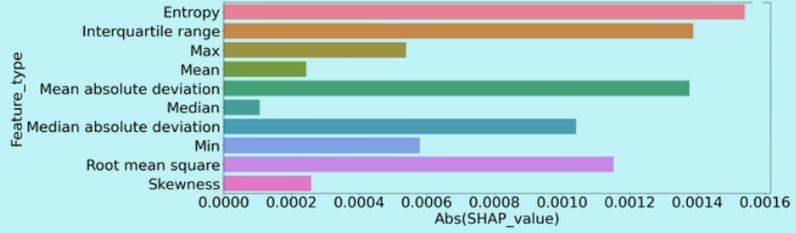
Extreme Gradient Boosting | SHAP

Results

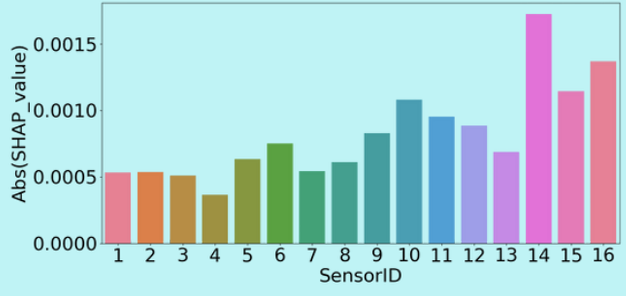


Original Matrix — Grad-CAM heatmap

Area of the most contribution sensors to the prediction



Most contribution features to the prediction



Most contribution sensors to the prediction